From GPS Feed to Performance Intelligence Layer

A proof of concept analysis using 7 years of personal Strava data

Donald Castellucci | Booth School of Business | BUSN 32120, Winter 2026

What is Strava?

180M+

Registered Athletes

Billions of

Activities Uploaded

#1

Endurance Sports App

Today, Strava is a GPS and social feed.

Athletes log their workouts (runs, rides, swims), share them with followers, and collect kudos.

But runners, cyclists, and triathletes are **the most data-obsessed people on the planet.**

Source: https://press.strava.com/articles/strava-releases-12th-annual-year-in-sport-trend-report-2025

Product Opportunities Unlock Premium Tier Growth

As we move toward IPO, analysts will look for product improvements that improve revenue and retention

01

Injury Prevention Leads to More Activities on App

Strava has every data point needed to flag dangerous training load spikes before they result in injury and keep athletes on the platform longer. Recent studies provide us the blueprint for a data-driven approach to injury prevention.

02

Weather and Conditions Data Improves Performance and Enjoyment

Surfacing weather data, including extreme condition moments and seasonal patterns creates shareable, create emotionally resonant athlete narratives along with opportunities for improved performance.

03

Engagement Intelligence and Year in Sport Additions Further Customer Stickiness

What drives kudos and comments? Understanding engagement unlocks smarter upload prompts, partner integrations, and subscription tier features. Additional Year in Sport features such as Top 10 Hottest Runs add new layer of content for users.

Single Athlete Proof of Concept Can Scale with Platform Data

7

Years of Data
2018 – 2026

4,945

Total Activities
Run, Ride & More

2,192

Runs Analyzed
After Cleaning

2

Data Sources
Strava + Weather

Data Sources

Strava API

4,945 activities via OAuth API pull. Includes distance, pace, heart rate, elevation, kudos, workout type, and social engagement data across 7 years.

Open-Meteo Historical Weather API

Hourly weather matched to each outdoor activity by GPS coordinates and start time. Temperature, humidity, wind speed, precipitation, and feels-like temperature.

Injury Prevention Tool Ensures Athletes Who Stay Healthy, Stay Engaged

The Framework

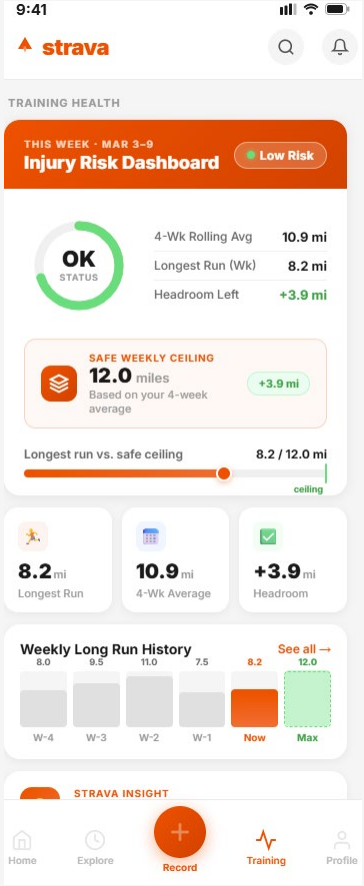
Rolling 4-week baseline.
Safe ceiling = 110% of prior average.
Flagged as OK, Exceeded, or Calibrating.

The Philosophy

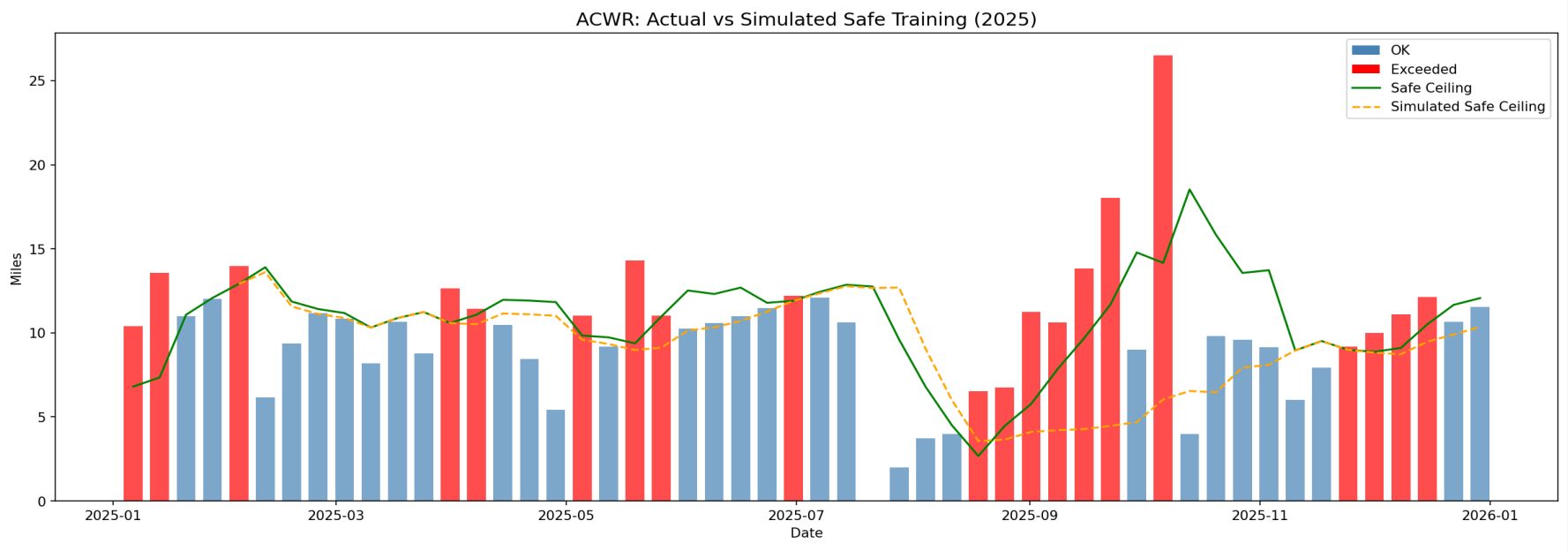
This doesn't stop athletes from running.
It gives them the intelligence to make informed decisions.

The Product

Every Monday morning, Strava sends:
your baseline, your safe ceiling this week,
your current risk status.



Acute Chronic Workload Dashboard in Action Reveals Challenges



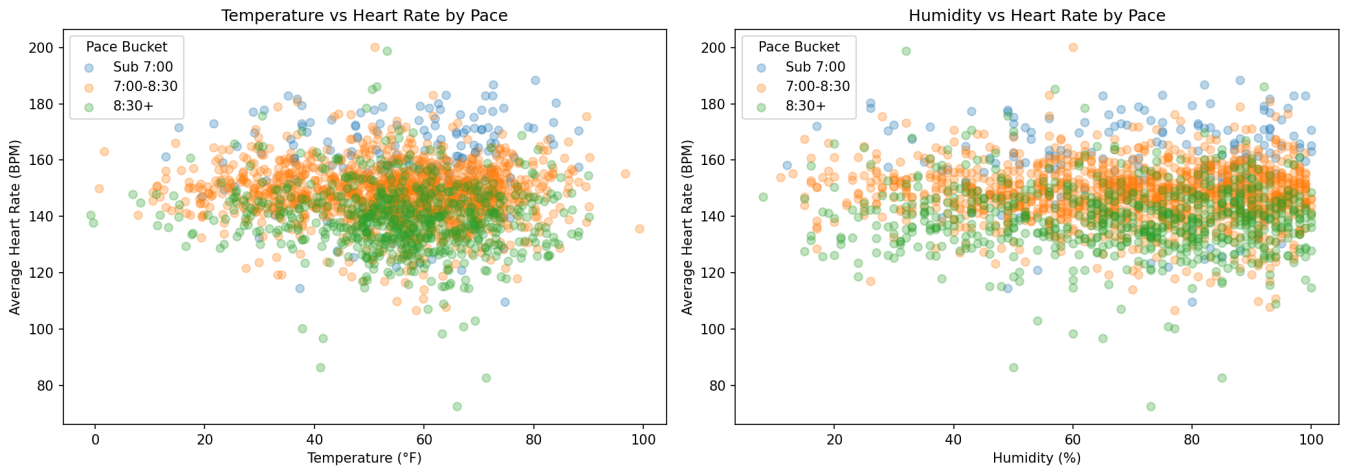
July Injury — Model Gave a Pass

The July 2025 injury was an isolated event as training load was actually within safe thresholds.

Chicago Marathon Build — The Real Risk

The September–October rebuild was the flagged period with an aggressive return-to-training after injury. The model warned for risk but the athlete ran through it and finished successfully.

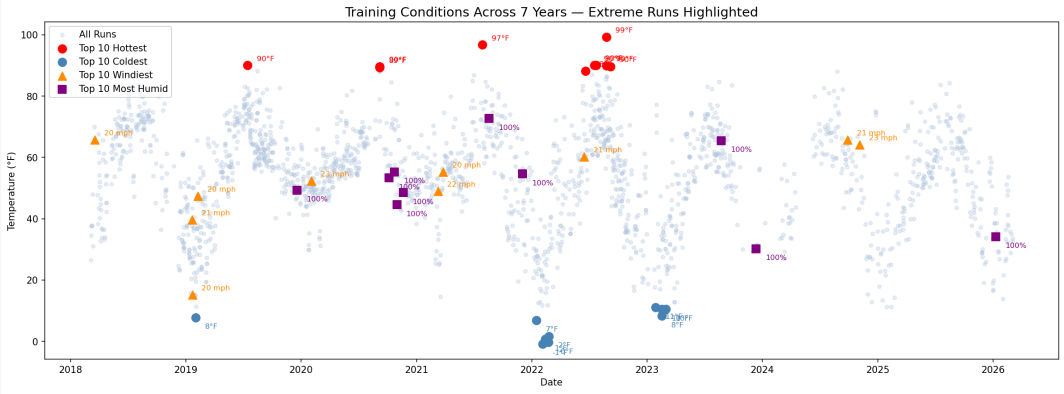
Null Result Turns Weather to Engagement tool, instead of Performance



More Data Needed for Stronger Performance Feature

Weather alone, even after controlling for pace, explained almost nothing about heart rate variation. We need sleep data, HRV, and recovery signals to close the gap. The data could come from partners or new internal hardware.

Opportunities Exist to Turn Weather Data into Engagement

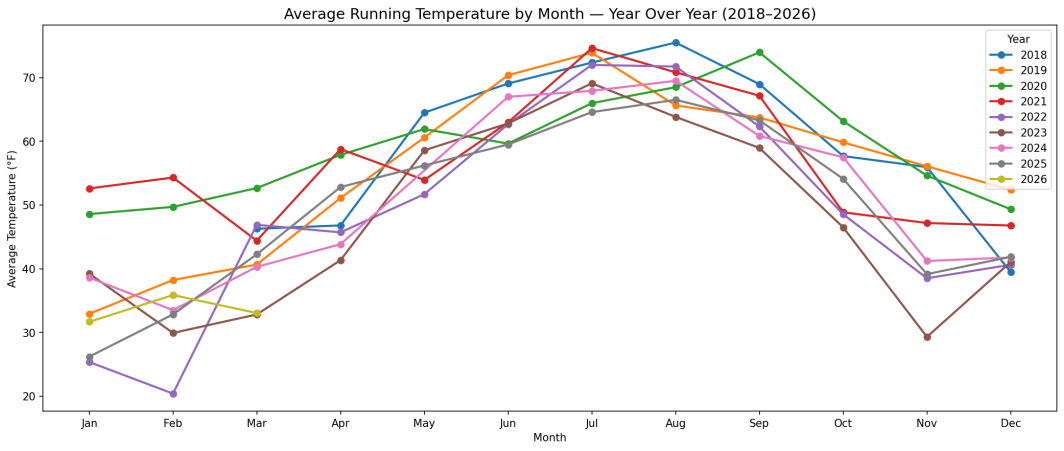


Weather Conditions Untapped for Year in Sport 2.0

"You ran in -1°F in February 2022"
"Your hottest run was 99°F in Las Vegas"

"Just wait for the summer, it'll be beautiful!" – everyone, here

Was it really way colder last winter? By connecting athlete activity data with corresponding weather, we create opportunities for users to engage with unique visuals.



Nudging Athletes to Display Personality through the Kudos Model

R^2

0.51

Model explains 51% of kudos variation

RMSE

6.5

Within ~6-7 kudos of actual count

MAE

~4

Avg error of 4 kudos per prediction

Top Engagement Drivers

1	**Distance**	Longer efforts earn more kudos
2	**Has Photo**	Directly controllable at upload
3	**Custom Title**	Directly controllable at upload
4	**Race / Long Run Tag**	Community rewards harder efforts

Product Recommendation

Strava already prompts photo uploads. Extend this to custom title prompts at the point of upload.

Two of the top three engagement drivers require simply a nudge at the right moment.

Top 5 Strava Activities by Engagement Driver

1	**Kudos: 95**	26.2	No	Yes	Yes
2	**Kudos: 84**	26.2	Yes	Yes	Yes
3	**Kudos: 79**	26.2	Yes	Yes	Yes
4	**Kudos: 75**	26.2	No	Yes	Yes
5	**Kudos: 65**	26.2	Yes	Yes	Yes
-	**Kudos: 5**	4.3	No	No	No

Product Roadmap

Four features that emerge directly from this analysis

01 ACWR Injury Risk Dashboard

Weekly training load status and safe long run ceiling surfaced natively. Natural home: Strava's Fitness & Freshness tab.

02 Year in Sport 2.0

Extreme condition badges, seasonal training narratives, and partner activation timing.

03 Engagement Prompts at Upload

Extend existing photo prompt to include custom title nudge.

04 Pre-Run Readiness Score

Composite score combining ACWR load, sleep, HRV, and weather. Requires device partnerships (Garmin, WHOOP) or internal hardware. The full intelligence layer vision.

Limitations & Future State

Stryd Running Power Data

Per-second power CSVs from a Stryd foot pod would replace pace as the effort metric because it would be more precise, less confounded by terrain and conditions. No public API.

Garmin Sleep & HRV Data

Sleep staging, HRV, and recovery scores likely account for the unexplained 70% of the heart rate model. No public API.

Platform Scale

Every model presented is a single-athlete proof of concept. At 180M users, the kudos model separates individual social network effects from true engagement drivers. The ACWR model builds population-level injury risk benchmarks. Pace zones were defined using domain knowledge for this proof of concept; at platform scale, Strava could use unsupervised clustering on each athlete's pace-HR relationship to generate personalized effort zones automatically.

Strava's moat?

Billions of activities worth of athlete intelligence that no competitor can replicate.

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github.com/dcastellucci8/strava-performance-intelligence